ADVENTURES IN BIG MATH · 3RD GRADE MATH

MISSION 06 OF 08 · WORKSHEET PACK

Geometry Quest

Every graded set, every week — 5 weeks, 25 two-page sheets, with an answer key after each week.

WEEK 1	5 sets
WEEK 2	5 sets
WEEK 3	5 sets
WEEK 4	5 sets
WEEK 5	5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

What an Angle Is

Every answer is a number of degrees, or a count. Type digits only.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

8 ITEMS · RECALL

1 Right angles in a full turn ____

2 Right angles in a straight line ____

3 Perimeter of a 5 by 2 rectangle ____

4 Perimeter of a 4 by 4 square ____

5 Perimeter of a 8 by 3 rectangle ____

6 Perimeter of a 7 by 7 square ____

7 Perimeter of a 9 by 1 rectangle ____

8 Perimeter of a 2 by 2 square ____

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

9. Perimeter of a 14 by 6 rectangle ....

10. A square of perimeter 24 — one side ....

11. A rectangle of perimeter 26, one side 8 — the other ....

12. Perimeter of a 20 by 11 rectangle ....

13. A square of perimeter 64 — one side ....

★ Challenge

NOT GRADED · REASON IT OUT

14. Degrees in a right angle
15. Degrees in a straight line
16. Degrees all the way round
17. Degrees in half a right angle
18. On a straight line: 130° and ?° (180 - 130)
19. Round a point: 90°, 120° and ?° (360 - 210)
20. Three angles of a triangle: 60°, 70° and ?° (They add to 180)
21. An angle of 200° — how far past a straight line? (200 - 180)
22. Two equal angles on a straight line — each is ?° (180 shared evenly)
23. Angles in a square, added up (Four right angles)
24. Turn 90° four times — total degrees (All the way round)
25. A rectangle of perimeter 24 with whole sides — how many different ones

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Perimeter & Area

Perimeter is the fence. Area is the grass. Read which one is being asked for.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

12 ITEMS · RECALL

1 Perimeter of a 5 by 3 rectangle	2 Area of a 5 by 3 rectangle	3 Perimeter of a square with side 6	4 Area of a square with side 6
5 Perimeter of a 10 by 2 rectangle	6 Area of a 10 by 2 rectangle	7 Area of a 5 by 2 rectangle	8 Area of a 4 by 4 square
9 Area of a 8 by 3 rectangle	10 Area of a 7 by 7 square	11 Area of a 9 by 1 rectangle	12 Area of a 6 by 5 rectangle

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Area 48, one side 6 — the other side (48 ÷ 6) ....
14. Perimeter 30, one side 9 — the other side (9 + 9 is 18, and 12 is left for two sides) ....
15. Area of a 12 by 7 rectangle (84 square units) ....
16. Perimeter of a 12 by 7 rectangle (12 + 12 + 7 + 7) ....
17. L-shape: a 6 by 4 block plus a 3 by 2 block — total area (24 + 6) ....
18. Area of a 14 by 6 rectangle ....
19. Area 60, one side 5 — the other ....
20. Area of a 20 by 11 rectangle ....
21. A square of area 81 — one side ....
22. Area 169 with equal sides — one side ....

★ Challenge

NOT GRADED · REASON IT OUT

23. Biggest area with perimeter 24 (whole sides) (The 6 by 6 square)

24. Smallest area with perimeter 24 (whole sides) (1 by 11)

25. Area of an L: a 7 by 7 square minus a 3 by 2 corner

26. Double both sides of a 3 by 6 rectangle — the new area

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

The Coordinate Plane

Across first, then up. (3, 5) is a different place from (5, 3).

Warm-Up

12 ITEMS · RECALL

1 In (3, 5), the across number	2 In (3, 5), the up number	3 In (7, 2), the across number	4 In (0, 4), the across number
5 In (6, 6), the up number	6 In (9, 1), the up number	7 A 1 by 9 rectangle — perimeter	8 That rectangle — area
9 A 5 by 5 square — perimeter	10 That square — area	11 A 2 by 8 rectangle — area	12 A 3 by 7 rectangle — area

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Distance from (2, 3) to (7, 3) (Only the across number changed) ....
14. Distance from (4, 1) to (4, 8) (8 - 1) ....
15. Rectangle at (1,1), (6,1), (6,4), (1,4) — its perimeter (5 wide and 3 tall) ....
16. That same rectangle — its area (5 × 3) ....
17. Start at (2,2), go 4 across and 3 up — the up number now (2 + 3) ....
18. With 20 m of fence, the biggest whole-number area ....
19. With 28 m of fence, the biggest area ....
20. With 12 m of fence, the biggest area ....
21. With 20 m of fence, the smallest area using whole sides ....
22. With 36 m of fence, the biggest whole-number area ....

★ Challenge

NOT GRADED · REASON IT OUT

23. Square with corners (0,0), (5,0), (5,5) — the fourth corner's up number (It sits at (0, 5))

24. Distance from (1,2) to (1,9) plus (3,4) to (8,4) (7 + 5)

25. Among rectangles with perimeter 32, the biggest area

26. With 100 m of fence, the biggest whole-number area

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Sorting Shapes & Symmetry

Sort by what is true about a shape, not by what it looks like.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

12 ITEMS · RECALL

1 Sides on a hexagon	2 Sides on a pentagon	3 Right angles in a rectangle	4 Equal sides on an equilateral triangle
5 Sides on an octagon	6 Pairs of parallel sides in a rectangle	7 A 4 by 2 plus a 4 by 2 — total area	8 A 5 by 5 plus a 3 by 3
9 A 6 by 3 plus a 6 by 1	10 A 7 by 2 plus a 4 by 2	11 A 12 by 1 plus a 6 by 2	12 A 8 by 3 plus a 2 by 3

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Lines of symmetry in a square (Two through the middle, two through the corners) ....

14. Lines of symmetry in a rectangle (not a square) (The corner folds don't match) ....

15. Lines of symmetry in an equilateral triangle (One from each corner) ....

16. Lines of symmetry in a circle — type 0 if too many to count (Infinitely many, so type 0) ....

17. A 9 by 7 with a 3 by 2 removed — area ....

18. A 12 by 12 with a 5 by 5 removed ....

19. An L of a 10 by 5 and a 6 by 4 ....

20. A 14 by 6 with a 4 by 6 removed ....

21. A T of a 10 by 2 and a 2 by 7 ....

★ Challenge

NOT GRADED · REASON IT OUT

22. Degrees in a quarter turn (A right angle)

23. Lines of symmetry in a regular hexagon (One per pair of opposite corners and sides)

24. Sides of a shape whose angles add to 360 (Any quadrilateral)

25. A 25 by 18 room with a 5 by 6 alcove added — total area

26. A 15 by 15 with a 15 by 4 strip removed

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

NAME __________

DATE __________

RIGHT ____ OF ____ TRIED

Angle Guess & Map a Planet

Enrichment day. Estimate before you measure, every single time.

Warm-Up

2 ITEMS · RECALL

1 With 20 m and no wall, the biggest area

2 Sides you must fence with a wall behind you

Core

GRADED · SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

3. 20 m against a wall, 5 m deep each side — the width

4. That pen's area

Challenge

NOT GRADED · REASON IT OUT

5. Angle on a straight line next to 47° (180 - 47)
6. Angles round a point: 100°, 130° and ?° (360 - 230)
7. Rectangle perimeter 20 with the largest area — that area (The 5 by 5 square)
8. Rectangle area 36 with the smallest perimeter — that perimeter (6 by 6)
9. Distance from (3,7) to (11,7) (11 - 3)
10. A shape with 4 lines of symmetry and 4 equal sides — its angle count (It's a square)
11. Turn 45° eight times — total degrees (45 × 8)
12. Interior angles of a triangle, added (Always)
13. 20 m against a wall, the biggest area
14. 32 m against a wall, the biggest area
15. How much bigger is that than the no-wall best for 32 m

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

TEACHER PAGE · ANSWER KEY

Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

1.1 · What an Angle Is

OUT OF 13

Warm-Up · 1–8 · 4 | 2 | 14 | 16 | 22 | 28 | 20 | 8

Core · 9–13 · 40 | 6 | 5 | 62 | 16

Challenge · 14–25 · 90 | 180 | 360 | 45 | 50 | 150 | 50 | 20 | 90 | 360 | 360 | 6

1.2 · Perimeter & Area

OUT OF 22

Warm-Up · 1–12 · 16 | 15 | 24 | 36 | 24 | 20 | 10 | 16 | 24 | 49 | 9 | 30

Core · 13–22 · 8 | 6 | 84 | 38 | 30 | 84 | 12 | 220 | 9 | 13

Challenge · 23–26 · 36 | 11 | 43 | 72

1.3 · The Coordinate Plane

OUT OF 22

Warm-Up · 1–12 · 3 | 5 | 7 | 0 | 6 | 1 | 20 | 9 | 20 | 25 | 16 | 21

Core · 13–22 · 5 | 7 | 16 | 15 | 5 | 25 | 49 | 9 | 9 | 81

Challenge · 23–26 · 5 | 12 | 64 | 625

1.4 · Sorting Shapes & Symmetry

OUT OF 21

Warm-Up · 1–12 · 6 | 5 | 4 | 3 | 8 | 2 | 16 | 34 | 24 | 22 | 24 | 30

Core · 13–21 · 4 | 2 | 3 | 0 | 57 | 119 | 74 | 60 | 34

Challenge · 22–26 · 90 | 6 | 4 | 480 | 165

Fri · Angle Guess & Map a Planet

OUT OF 4

Warm-Up · 1–2 · 25 | 3

Core · 3–4 · 10 | 50

Challenge · 5–15 · 133 | 130 | 25 | 24 | 8 | 4 | 360 | 180 | 50 | 128 | 64

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Perimeter

The distance all the way round. Add every side, or be clever about it.

Warm-Up 12 ITEMS · RECALL

1 Perimeter of a 4 by 3 rectangle	2 Perimeter of a 5 by 5 square	3 Perimeter of a 10 by 2 rectangle	4 Perimeter of a 6 by 6 square
5 Perimeter of a 7 by 1 rectangle	6 Perimeter of a 3 by 3 square	7 Perimeter of a 6 by 2 rectangle	8 Perimeter of a 8 by 8 square
9 Perimeter of a 12 by 1 rectangle	10 Perimeter of a 9 by 9 square	11 Perimeter of a 5 by 4 rectangle	12 Perimeter of a 10 by 10 square

Core GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Perimeter of a 12 by 8 rectangle ....
14. A square of perimeter 36 — one side ....
15. A rectangle of perimeter 20, one side 6 — the other ....
16. Perimeter of a 15 by 9 rectangle ....
17. A square of perimeter 100 — one side ....
18. Perimeter of a 16 by 9 rectangle ....
19. A square of perimeter 48 — one side ....
20. A rectangle of perimeter 34, one side 10 — the other ....
21. Perimeter of a 25 by 15 rectangle ....
22. A square of perimeter 200 — one side ....

Challenge

NOT GRADED · REASON IT OUT

23. A rectangle of perimeter 30 with whole sides — how many different ones

24. Perimeter of an L made from a 5 by 5 square with a 2 by 2 corner removed

25. A rectangle of perimeter 40 with whole sides — how many different ones

26. Perimeter of an L from a 8 by 8 square with a 3 by 3 corner removed

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Area

Length times width — and count the squares if you doubt it.

Warm-Up

12 ITEMS · RECALL

1 Area of a 4 by 3 rectangle

2 Area of a 5 by 5 square

3 Area of a 10 by 2 rectangle

4 Area of a 6 by 6 square

5 Area of a 7 by 1 rectangle

6 Area of a 8 by 3 rectangle

7 Area of a 6 by 2 rectangle

8 Area of a 8 by 8 square

9 Area of a 12 by 1 rectangle

10 Area of a 9 by 9 square

11 Area of a 5 by 4 rectangle

12 Area of a 10 by 7 rectangle

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Area of a 12 by 8 rectangle
14. Area 48, one side 6 — the other
15. Area of a 15 by 9 rectangle
16. A square of area 49 — one side
17. Area 144 with equal sides — one side
18. Area of a 16 by 9 rectangle
19. Area 72, one side 8 — the other
20. Area of a 25 by 15 rectangle
21. A square of area 100 — one side
22. Area 225 with equal sides — one side

★ Challenge

NOT GRADED · REASON IT OUT

23. Area of an L: a 6 by 6 square minus a 2 by 3 corner

24. Double both sides of a 4 by 5 rectangle — the new area

25. Area of an L: a 8 by 8 square minus a 3 by 4 corner

26. Double both sides of a 5 by 7 rectangle — the new area

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Same Fence, Different Field

Perimeter fixed, area wildly different. That is the Big Question.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

12 ITEMS · RECALL

1 A 1 by 11 rectangle — perimeter

2 That rectangle — area

3 A 6 by 6 square — perimeter

4 That square — area

5 A 2 by 10 rectangle — area

6 A 4 by 8 rectangle — area

7 A rectangle cut into 2 equal parts — one part is 1 over what

8 A rectangle cut into 3 equal parts — one part is 1 over what

9 A rectangle cut into 4 equal parts — one part is 1 over what

10 A rectangle cut into 6 equal parts — one part is 1 over what

11 A rectangle cut into 8 equal parts — one part is 1 over what

12 A 1 by 15 rectangle — perimeter

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

18. With 24 m of fence, the biggest whole-number area
19. With 20 m of fence, the biggest area
20. With 16 m of fence, the biggest area
21. With 24 m of fence, the smallest area using whole sides
22. With 30 m of fence, the biggest whole-number area
23. A shape of 8 squares split into 2 equal parts — squares in each part
24. A shape of 12 squares split into 4 equal parts — squares in each part
25. A shape of 9 squares split into 3 equal parts — squares in each part
26. A shape of 18 squares split into 6 equal parts — squares in each part
27. A square folded into 4 equal parts — each part is what fraction, give the denominator
28. Two of four equal parts — the same as one out of how many

CHALLENGE AND THE ERROR JOURNAL ARE ON THE BACK

Challenge

NOT GRADED · REASON IT OUT

34. The shape that always gives the biggest area for a fixed perimeter — type square or circle

35. Among rectangles with perimeter 40, the biggest area

36. A 12-square shape split into 3 equal parts, then one part split in 2 — squares in the smallest piece

37. The shape giving the biggest area for a fixed perimeter — type square or circle

38. Among rectangles with perimeter 60, the biggest area

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

Area of Compound Shapes

Split it into rectangles, do each, add them up.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

12 ITEMS · RECALL

1 A 3 by 2 plus a 3 by 2 — total area

2 A 4 by 4 plus a 2 by 2

3 A 5 by 3 plus a 5 by 1

4 A 6 by 2 plus a 3 by 2

5 A 10 by 1 plus a 5 by 2

6 A 7 by 3 plus a 1 by 3

7 A 4 by 3 plus a 4 by 3 — total area

8 A 6 by 6 plus a 3 by 3

9 A 7 by 4 plus a 7 by 1

10 A 9 by 2 plus a 4 by 2

11 A 15 by 1 plus a 5 by 3

12 A 6 by 4 plus a 2 by 4

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. An 8 by 6 with a 2 by 3 removed — area
14. A 10 by 10 with a 4 by 4 removed
15. An L of a 9 by 4 and a 5 by 3
16. A 12 by 5 with a 3 by 5 removed
17. A T of a 8 by 2 and a 2 by 6
18. A 10 by 8 with a 3 by 4 removed — area
19. A 14 by 14 with a 6 by 6 removed
20. An L of a 12 by 5 and a 7 by 4
21. A 16 by 7 with a 5 by 7 removed
22. A T of a 12 by 3 and a 3 by 8

CHALLENGE AND THE ERROR JOURNAL ARE ON THE BACK

★ Challenge

NOT GRADED · REASON IT OUT

23. A 20 by 15 room with a 4 by 5 alcove added — total area
24. A 12 by 12 with a 12 by 3 strip removed
25. A 30 by 20 room with a 6 by 5 alcove added — total area
26. A 20 by 20 with a 20 by 6 strip removed

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

The Biggest Pen

24 metres of fence, and a wall to build against. Now what?

Warm-Up

4 ITEMS · RECALL

1 With 24 m and no wall, the biggest area

2 Sides you must fence with a wall behind you

3 With 36 m and no wall, the biggest area

4 Fenced sides with a wall behind

Core

GRADED · SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. 24 m against a wall, 8 m deep each side — the width ....

6. That pen's area ....

7. 36 m against a wall, 9 m deep each side — the width ....

8. 36 m against a wall, 9 m deep each side — area ....

Challenge

NOT GRADED · REASON IT OUT

9. 24 m against a wall, 6 m deep each side — the area

10. The biggest area for 24 m against a wall

11. How much bigger than the no-wall best

12. 36 m against a wall, the biggest area

13. 40 m against a wall, the biggest area

14. How much bigger than the no-wall best for 40 m

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

TEACHER PAGE · ANSWER KEY

Week 2 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

2.1 · Perimeter

OUT OF 22

Warm-Up · 1–12 · 14 | 20 | 24 | 24 | 16 | 12 | 16 | 32 | 26 | 36 | 18 | 40

Core · 13–22 · 40 | 9 | 4 | 48 | 25 | 50 | 12 | 7 | 80 | 50

Challenge · 23–26 · 7 | 20 | 10 | 32

2.2 · Area

OUT OF 22

Warm-Up · 1–12 · 12 | 25 | 20 | 36 | 7 | 24 | 12 | 64 | 12 | 81 | 20 | 70

Core · 13–22 · 96 | 8 | 135 | 7 | 12 | 144 | 9 | 375 | 10 | 15

Challenge · 23–26 · 30 | 80 | 52 | 140

2.3 · Same Fence, Different Field

OUT OF 23

Warm-Up · 1–12 · 24 | 11 | 24 | 36 | 20 | 32 | 2 | 3 | 4 | 6 | 8 | 32

Core · 18–28 · 36 | 25 | 16 | 11 | 56 | 4 | 3 | 3 | 3 | 4 | 2

Challenge · 34–38 · circle | 100 | 2 | circle | 225

2.4 · Area of Compound Shapes

OUT OF 22

Warm-Up · 1–12 · 12 | 20 | 20 | 18 | 20 | 24 | 24 | 45 | 35 | 26 | 30 | 32

Core · 13–22 · 42 | 84 | 51 | 45 | 28 | 68 | 160 | 88 | 77 | 60

Challenge · 23–26 · 320 | 108 | 630 | 280

Fri · The Biggest Pen

OUT OF 7

Warm-Up · 1–4 · 36 | 3 | 81 | 3

Core · 5–7 · 8 | 64 | 18

Challenge · 8–13 · 72 | 72 | 36 | 162 | 200 | 100

Plotting Points

Across the hall, then up the stairs. Type coordinates like 3,5.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

12 ITEMS · RECALL

1 In (3, 5), the number across	2 In (3, 5), the number up	3 The origin — type as a,b	4 4 across, 2 up — type as a,b
5 0 across, 6 up — type as a,b	6 In (7, 1), the first number	7 In (5, 2), the number across	8 In (5, 2), the number up
9 6 across, 0 up — type as a,b	10 2 across, 7 up — type as a,b	11 0 across, 9 up — type as a,b	12 In (4, 8), the first number

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 3 right of (2,5) — type as a,b
14. 2 up from (4,1) — type as a,b
15. Distance from (2,3) to (7,3)
16. Distance from (4,1) to (4,8)
17. Halfway between (0,0) and (8,0) — type as a,b
18. 5 right of (3,4) — type as a,b
19. 3 up from (6,2) — type as a,b
20. Distance from (1,5) to (9,5)
21. Distance from (7,2) to (7,10)
22. Halfway between (0,0) and (12,0) — type as a,b

CHALLENGE AND THE ERROR JOURNAL ARE ON THE BACK

Challenge

NOT GRADED · REASON IT OUT

23. Halfway between (2,2) and (8,6) — type as a,b
24. From (0,0) to (6,8) going only across and up — total steps
25. Halfway between (1,3) to (9,7) — type as a,b
26. From (0,0) to (7,9) going only across and up — total steps

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

Shapes on the Grid

Plot the corners, then read off what you built.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

12 ITEMS · RECALL

1 Corners on a rectangle	2 Corners on a triangle	3 (1,1) to (5,1) — the length	4 (1,1) to (1,4) — the length
5 Sides on a square	6 (0,0) to (3,0) — the length	7 Corners on a square	8 Corners on a pentagon
9 (2,1) to (7,1) — the length	10 (2,1) to (2,6) — the length	11 Sides on a rectangle	12 (0,0) to (5,0) — the length

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. (1,1), (1,5), (6,5), (6,1) — the area ....
14. (2,2),(2,6),(7,6),(7,2) — perimeter ....
15. (2,2), (2,7), (5,7), (5,2) — the area ....
16. (0,0), (0,4), (4,4), (4,0) — the shape's name ....
17. The missing corner of a rectangle at (1,1),(1,4),(6,4) — type as a,b ....
18. (1,1), (1,6), (5,6), (5,1) — the area ....
19. That shape's perimeter ....
20. (3,2), (3,8), (7,8), (7,2) — the area ....
21. (0,0), (0,6), (6,6), (6,0) — the shape's name ....
22. The missing corner of a rectangle at (2,2),(2,7),(8,7) — type as a,b ....

Challenge

NOT GRADED · REASON IT OUT

23. (0,0), (8,0), (8,5), (0,5) — the area

24. A square with corners (2,2) and (7,7) opposite — its area

25. (0,0), (10,0), (10,6), (0,6) — the area

26. A square with corners (1,1) and (8,8) opposite — its area

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

NAME __________

DATE __________

RIGHT ____ OF ____ TRIED

Reading a Map

Coordinates are directions somebody else has to follow.

Warm-Up

12 ITEMS · RECALL

1 From (0,0), move 3 across — type as a,b

2 From (3,0), move 2 up — type as a,b

3 From (5,5), move 1 across — type as a,b

4 From (2,2), move 2 up — type as a,b

5 Blocks from (0,0) to (4,0)

6 Blocks from (0,0) to (0,7)

7 From (0,0), move 4 across — type as a,b

8 From (4,0), move 3 up — type as a,b

9 From (6,6), move 2 across — type as a,b

10 From (3,3), move 4 up — type as a,b

11 Blocks from (0,0) to (6,0)

12 Blocks from (0,0) to (0,9)

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. From (1,2) to (6,2) then up 3 — type the end as a,b

14. Total blocks travelled on that route

15. A park at (3,4), a school at (9,4) — blocks apart

16. From (2,1) to (2,9) — blocks

17. From (0,0) to (5,5) across-and-up — blocks

18. From (2,3) to (8,3) then up 4 — type the end as a,b

19. From (2,3) to (8,3) then up 4 — total blocks travelled

20. A park at (2,5), a school at (11,5) — blocks apart

21. From (3,2) to (3,11) — blocks

22. From (0,0) to (6,6) across-and-up — blocks

Challenge

NOT GRADED · REASON IT OUT

23. A route (1,1) to (7,1) to (7,6) — total blocks

24. The shortest across-and-up route from (2,3) to (9,8) — blocks

25. A route (2,2) to (9,2) to (9,8) — total blocks

26. The shortest across-and-up route from (1,2) to (10,9) — blocks

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

Order Matters

(3,5) and (5,3) are different places. Prove it every time.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

12 ITEMS · RECALL

1 In (3,5), the across value

2 In (5,3), the across value

3 Are (3,5) and (5,3) the same — yes or no

4 In (0,4), the across value

5 In (4,0), the up value

6 A point on the bottom line has which value 0 — type across or up

7 In (4,7), the across value

8 In (7,4), the across value

9 Are (4,7) and (7,4) the same — yes or no

10 In (0,5), the across value

11 In (5,0), the up value

12 A point on the left edge has which value 0 — type across or up

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. (2,6) and (6,2) — how far apart across
14. Those two points — how far apart up
15. A point with the same across and up as (5,5) — type as a,b
16. Points where across equals up, from (0,0) to (5,5) — how many
17. Swap (7,2) — type the new point as a,b
18. (3,8) and (8,3) — how far apart across
19. (3,8) and (8,3) — how far apart up
20. A point with the same across and up as (7,7) — type as a,b
21. Points where across equals up, from (0,0) to (7,7) — how many
22. Swap (9,4) — type the new point as a,b

★ Challenge

NOT GRADED · REASON IT OUT

23. Points on the line where across equals up, from 0 to 10 — how many

24. (1,4), (4,1), (1,1) — the area of that triangle

25. Points on the line where across equals up, from 0 to 20 — how many

26. (2,5), (5,2), (2,2) — the area of that triangle

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

4 ITEMS · RECALL

1 Perimeter of a 4 by 3 rectangle

2 Area of a 4 by 3 rectangle

3 Perimeter of a 5 by 4 rectangle

4 Area of a 5 by 4 rectangle

Core

GRADED · SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. Area 48, one side 6 — the other
6. With 20 m of fence, the biggest area
7. An 8 by 6 with a 2 by 3 removed — area
8. Distance from (2,3) to (7,3)
9. (1,1), (1,5), (6,5), (6,1) — the area
10. Area 72, one side 8 — the other
11. With 40 m of fence, the biggest area
12. A 10 by 8 with a 3 by 4 removed — area
13. Distance from (1,5) to (9,5)
14. (1,1), (1,6), (5,6), (5,1) — the area

★ Challenge

NOT GRADED · REASON IT OUT

15. A rectangle of perimeter 30 with whole sides — how many different ones

16. A rectangle of perimeter 40 with whole sides — how many different ones

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

TEACHER PAGE · ANSWER KEY

Week 3 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

3.1 · Plotting Points

OUT OF 22

Warm-Up · 1–12 · 3 | 5 | 0,0 | 4,2 | 0,6 | 7 | 5 | 2 | 6,0 | 2,7 | 0,9 | 4

Core · 13–22 · 5,5 | 4,3 | 5 | 7 | 4,0 | 8,4 | 6,5 | 8 | 8 | 6,0

Challenge · 23–26 · 5,4 | 14 | 5,5 | 16

3.2 · Shapes on the Grid

OUT OF 22

Warm-Up · 1–12 · 4 | 3 | 4 | 3 | 4 | 3 | 4 | 5 | 5 | 5 | 4 | 5

Core · 13–22 · 20 | 18 | 15 | square | 6,1 | 20 | 18 | 24 | square | 8,2

Challenge · 23–26 · 40 | 25 | 60 | 49

3.3 · Reading a Map

OUT OF 22

Warm-Up · 1–12 · 3,0 | 3,2 | 6,5 | 2,4 | 4 | 7 | 4,0 | 4,3 | 8,6 | 3,7 | 6 | 9

Core · 13–22 · 6,5 | 8 | 6 | 8 | 10 | 8,7 | 10 | 9 | 9 | 12

Challenge · 23–26 · 11 | 12 | 13 | 16

3.4 · Order Matters

OUT OF 22

Warm-Up · 1–12 · 3 | 5 | no | 0 | 0 | up | 4 | 7 | no | 0 | 0 | across

Core · 13–22 · 4 | 4 | 5,5 | 6 | 2,7 | 5 | 5 | 7,7 | 8 | 4,9

Challenge · 23–26 · 11 | 4.5 | 21 | 4.5

Fri · Mid-Unit Quiz

OUT OF 14

Warm-Up · 1–4 · 14 | 12 | 18 | 20

Core · 5–14 · 8 | 25 | 42 | 5 | 20 | 9 | 100 | 68 | 8 | 20

Challenge · 15–16 · 7 | 10

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Lines of Symmetry

Fold it. If both halves land exactly, that fold is a line of symmetry.

Warm-Up

11 ITEMS · RECALL

1 Lines of symmetry in a square

2 In a rectangle that is not a square

3 In a circle — type the word

4 In an equilateral triangle

5 In the letter A

6 In the letter H

7 Lines of symmetry in a rectangle

8 In the letter O — type the word

9 In the letter T

10 In the letter X

11 In a square

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. In a regular pentagon ....

13. In a regular hexagon ....

14. In an isosceles triangle ....

15. In a scalene triangle ....

16. In a rhombus that is not a square ....

17. In a regular heptagon ....

18. In a regular nonagon ....

19. In a kite ....

20. In a parallelogram that is not a rhombus ....

21. In a regular decagon ....

Challenge

NOT GRADED · REASON IT OUT

22. In a regular octagon

23. In a regular polygon with 12 sides

24. In a regular polygon with 20 sides

25. In a semicircle

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

Flips and Turns

Slide, flip, turn. The shape keeps its size and its angles.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

4 ITEMS · RECALL

1 Quarter turns in a full turn

2 Half turns in a full turn

3 Quarter turns in a half turn

4 Half turns in two full turns

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

5. (2,3) flipped over the vertical line through 0 — the across value becomes

6. Slide (2,3) four right — type as a,b

7. (3,4) flipped over the vertical line through 0 — the across value becomes

8. Slide (3,4) five up — type as a,b

★ Challenge

NOT GRADED · REASON IT OUT

9. Degrees in a quarter turn
10. Degrees in a half turn
11. Degrees in a full turn
12. Degrees in three quarter turns
13. A square turned 90° looks the same — yes or no
14. Turns of 90° before a square returns to start
15. A rectangle looks the same after how many degrees
16. An equilateral triangle looks the same after how many degrees
17. A regular hexagon — the smallest turn that leaves it unchanged, in degrees
18. Degrees in two right angles
19. Degrees in a three-quarter turn
20. An equilateral triangle turned 120° looks the same — yes or no

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Angles

Right, acute, obtuse — and what they add to.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up 12 ITEMS · RECALL

1 Angles in a triangle add to

2 A 4 by 3 rectangle joined to a 4 by 2 — total squares

3 A 5 by 2 rectangle joined to a 5 by 4 — total squares

4 A 6 by 3 rectangle joined to a 6 by 3 — total squares

5 A 3 by 5 rectangle joined to a 3 by 5 — total squares

6 A 7 by 2 rectangle joined to a 7 by 3 — total squares

7 A rectangle 3 squares across and 4 down — squares altogether

8 A rectangle 5 squares across and 2 down — squares altogether

9 A rectangle 6 squares across and 3 down — squares altogether

10 A rectangle 4 squares across and 4 down — squares altogether

11 A rectangle 7 squares across and 2 down — squares altogether

12 A rectangle 5 squares across and 5 down — squares altogether

Core GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Angles in a quadrilateral add to

14. 4 × (3 + 2) — work out the bracket first

15. 4 × 3 plus 4 × 2

16. 5 × (2 + 4) — work out the bracket first

17. 5 × 2 plus 5 × 4

18. 6 × (3 + 3) — work out the bracket first

19. 6 × 3 plus 6 × 3

20. A 8 by 3 rectangle and a 8 by 5 joined — total area

21. Split a 7 by 6 rectangle into 7 by 5 and 7 by 1 — total area

22. A 9 by 3 rectangle and a 9 by 4 joined — total area

23. An L shape: a 4 by 3 rectangle and a 2 by 3 rectangle. Total area

CHALLENGE AND THE ERROR JOURNAL ARE ON THE BACK

Challenge

NOT GRADED · REASON IT OUT

43. Degrees in a right angle
44. Degrees on a straight line
45. Degrees round a point
46. An angle of 45° — type acute or obtuse
47. An angle of 120° — acute or obtuse
48. Two angles on a line, one is 60° — the other
49. A triangle with 90° and 30° — the third angle
50. A quadrilateral with three 90° angles — the fourth
51. Three angles round a point, two are 100° — the third
52. Each angle of an equilateral triangle
53. Each angle of a regular hexagon
54. A 10 by 7 rectangle with a 3 by 2 corner removed — area left

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

Classifying Shapes

Sides, angles, parallels. Name it from its properties.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

✦ Warm-Up

12 ITEMS · RECALL

1 Sides on a pentagon	2 Sides on a hexagon	3 Sides on an octagon	4 Sides on a quadrilateral
5 Equal sides on an equilateral triangle	6 Sides on a triangle	7 Sides on a heptagon	8 Sides on a nonagon
9 Sides on a decagon	10 Sides on a rhombus	11 Equal sides on an isosceles triangle	12 Sides on a square

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Is every square a rectangle — yes or no ....
14. Is every rectangle a square ....
15. Pairs of parallel sides in a parallelogram ....
16. Pairs of parallel sides in a trapezoid ....
17. Is every rhombus a parallelogram — yes or no ....
18. Is every parallelogram a rhombus ....
19. Pairs of parallel sides in a rectangle ....
20. Right angles in a rectangle ....

Challenge

NOT GRADED · REASON IT OUT

21. A triangle with one 90° angle — type its name

22. A rhombus with right angles is also called a

23. A triangle with all sides different — type its name

24. A triangle with all angles under 90° — type its name

25. A parallelogram with four equal sides is also called a

26. A triangle with exactly two equal sides — type its name

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Tessellation

Which shapes tile a floor with no gaps? Test three and explain.

Warm-Up

4 ITEMS · RECALL

1 Do squares tile with no gaps — yes or no

2 Do circles tile with no gaps

3 Do rectangles tile with no gaps — yes or no

4 Do regular octagons tile alone

Core

GRADED · SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. Do equilateral triangles tile — yes or no

6. Do regular hexagons tile

7. Do parallelograms tile — yes or no

8. Do regular heptagons tile

Challenge

NOT GRADED · REASON IT OUT

9. Do regular pentagons tile — yes or no

10. Angles must meet at a point adding to

11. Hexagons at a point: 360 ÷ 120 — how many meet

12. Squares at a point: 360 ÷ 90 — how many meet

13. Triangles at a point: 360 ÷ 60 — how many meet

14. Why pentagons fail: 360 ÷ 108 is not a whole number — type the remainder to the nearest whole degree

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

TEACHER PAGE · ANSWER KEY

Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 · Lines of Symmetry

OUT OF 21

Warm-Up · 1–11 · 4 | 2 | infinite | 3 | 1 | 2 | 2 | infinite | 1 | 2 | 4

Core · 12–21 · 5 | 6 | 1 | 0 | 2 | 7 | 9 | 1 | 0 | 10

Challenge · 22–25 · 8 | 12 | 20 | 1

4.2 · Flips and Turns

OUT OF 8

Warm-Up · 1–4 · 4 | 2 | 2 | 4

Core · 5–8 · -2 | 6,3 | -3 | 3,9

Challenge · 9–20 · 90 | 180 | 360 | 270 | yes | 4 | 180 | 120 | 60 | 180 | 270 | yes

4.3 · Angles

OUT OF 23

Warm-Up · 1–12 · 180 | 20 | 30 | 36 | 30 | 35 | 12 | 10 | 18 | 16 | 14 | 25

Core · 13–23 · 360 | 20 | 20 | 30 | 30 | 36 | 36 | 64 | 42 | 63 | 18

Challenge · 43–54 · 90 | 180 | 360 | acute | obtuse | 120 | 60 | 90 | 160 | 60 | 120 | 64

4.4 · Classifying Shapes

OUT OF 20

Warm-Up · 1–12 · 5 | 6 | 8 | 4 | 3 | 3 | 7 | 9 | 10 | 4 | 2 | 4

Core · 13–20 · yes | no | 2 | 1 | yes | no | 2 | 4

Challenge · 21–26 · right | square | scalene | acute | rhombus | isosceles

Fri · Tessellation

OUT OF 8

Warm-Up · 1–4 · yes | no | yes | no

Core · 5–8 · yes | yes | yes | no

Challenge · 9–14 · no | 360 | 3 | 4 | 6 | 36

Map a Planet

An invented world on the coordinate plane, with real distances.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

11 ITEMS · RECALL

1 A harbour at (2,3) — the across value

2 A peak at (7,7) — the up value

3 Distance from (2,3) to (6,3)

4 Distance from (5,1) to (5,6)

5 The origin — type as a,b

6 3 across, 8 up — type as a,b

7 A dock at (3,4) — the across value

8 A peak at (8,9) — the up value

9 Distance from (3,4) to (8,4)

10 Distance from (6,2) to (6,9)

11 5 across, 5 up — type as a,b

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Sailing (1,1) to (9,1) then to (9,5) — total distance

13. A square island (2,2) to (8,8) — its area

14. That island's perimeter

15. Halfway between (0,0) and (10,4) — type as a,b

16. A route (0,0) to (4,0) to (4,7) — distance

17. Sailing (2,2) to (10,2) then to (10,7) — total distance

18. A square island (1,1) to (9,9) — its area

19. A square island (1,1) to (9,9) — perimeter

20. Halfway between (0,0) and (12,8) — type as a,b

21. A route (1,1) to (6,1) to (6,9) — distance

★ Challenge

NOT GRADED · REASON IT OUT

22. A rectangular sea from (1,1) to (11,6) — its area

23. Ten landmarks each 3 units apart in a line — total length

24. A rectangular sea from (2,2) to (14,8) — its area

25. Twelve landmarks each 4 units apart in a line — total length

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Sailing Directions

Somebody else has to follow them. Ambiguity is the enemy.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up
12 ITEMS · RECALL

1 From (0,0) move 5 across — type as a,b

2 Then 3 up — type as a,b

3 Total distance travelled

4 From (4,4) move 2 up — type as a,b

5 From (4,4) move 2 across — type as a,b

6 Blocks from (0,0) to (3,4) across-and-up

7 From (0,0) move 7 across — type as a,b

8 Then 2 up — type as a,b

9 3 right then 2 up — total distance travelled

10 From (5,5) move 3 up — type as a,b

11 From (5,5) move 3 across — type as a,b

12 Blocks from (0,0) to (5,6) across-and-up

Core
GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. (1,1) to (1,7) to (5,7) — total distance ....

14. (2,2) to (8,2) to (8,9) — total distance ....

15. A round trip (0,0) to (6,0) and back — distance ....

16. (3,3) to (3,10) — distance ....

17. A square patrol of side 5 — total distance ....

18. (2,1) to (2,9) to (7,9) — total distance ....

19. (3,3) to (11,3) to (11,10) — total distance ....

20. A round trip (0,0) to (8,0) and back — distance ....

21. (4,4) to (4,12) — distance ....

22. A square patrol of side 7 — total distance ....

Challenge

NOT GRADED · REASON IT OUT

23. A patrol (1,1),(7,1),(7,5),(1,5) and back to start — total distance

24. The area enclosed by that patrol

25. A patrol (2,2),(9,2),(9,7),(2,7) and back — total distance

26. A patrol (2,2),(9,2),(9,7),(2,7) and back — the area enclosed

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Mission Review

Everything from five weeks, mixed together.

Warm-Up

10 ITEMS · RECALL

1 Perimeter of a 5 by 5 square ______

2 Area of a 5 by 5 square ______

3 Lines of symmetry in a square ______

4 In (3,5), the up value ______

5 Angles in a triangle add to ______

6 Perimeter of a 6 by 6 square ______

7 Area of a 6 by 6 square ______

8 Lines of symmetry in an equilateral triangle ______

9 In (6,2), the up value ______

10 Angles in a quadrilateral add to ______

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. Area 96 with one side 12 — the other ......

12. With 24 m of fence, the biggest area ......

13. (2,2), (2,7), (5,7), (5,2) — the area ......

14. Lines of symmetry in a regular hexagon ......

15. Area 144 with one side 16 — the other ......

16. With 32 m of fence, the biggest area ......

17. (3,2), (3,8), (7,8), (7,2) — the area ......

18. Lines of symmetry in a regular octagon ......

★ Challenge

NOT GRADED · REASON IT OUT

19. Degrees in a right angle

20. A triangle with 90° and 30° — the third angle

21. A 10 by 10 with a 4 by 4 removed — area

22. Each angle of a regular hexagon

23. Degrees on a straight line

24. A triangle with 50° and 60° — the third angle

25. A 14 by 14 with a 6 by 6 removed — area

26. Each angle of a regular pentagon

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

10 ITEMS · RECALL

1 Perimeter of a 6 by 6 square

2 Area of a 8 by 3 rectangle

3 Sides on a hexagon

4 Distance from (2,3) to (7,3)

5 Lines of symmetry in a rectangle

6 Perimeter of a 8 by 8 square

7 Area of a 9 by 4 rectangle

8 Sides on a decagon

9 Distance from (1,5) to (9,5)

10 Lines of symmetry in a square

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. An 8 by 6 with a 2 by 3 removed — area
12. A square of perimeter 36 — one side
13. (1,1), (1,5), (6,5), (6,1) — the perimeter
14. Is every square a rectangle — yes or no
15. A 10 by 8 with a 3 by 4 removed — area
16. A square of perimeter 48 — one side
17. (1,1), (1,6), (5,6), (5,1) — the perimeter
18. Is every rhombus a parallelogram — yes or no

CHALLENGE AND THE ERROR JOURNAL ARE ON THE BACK

Challenge

NOT GRADED · REASON IT OUT

19. Degrees round a point
20. Two angles on a line, one is 60° — the other
21. Among rectangles with perimeter 40, the biggest area
22. A regular hexagon — smallest turn leaving it unchanged, in degrees
23. Degrees in a full turn
24. Two angles on a line, one is 35° — the other
25. Among rectangles with perimeter 60, the biggest area
26. A regular octagon — smallest turn leaving it unchanged, in degrees

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

Mission 06 Test

Twelve items plus the Big Question, answered out loud.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

4 ITEMS · RECALL

1 Perimeter of a 10 by 2 rectangle

2 Area of a 10 by 2 rectangle

3 Perimeter of a 12 by 1 rectangle

4 Area of a 12 by 1 rectangle

Core

GRADED · SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. Area of a 12 by 8 rectangle
6. A rectangle of perimeter 20, one side 6 — the other
7. With 20 m of fence, the biggest area
8. A 12 by 5 with a 3 by 5 removed — area
9. Distance from (4,1) to (4,8)
10. (1,1), (1,5), (6,5), (6,1) — the area
11. Lines of symmetry in a regular pentagon
12. Area of a 16 by 9 rectangle
13. A rectangle of perimeter 26, one side 8 — the other
14. With 40 m of fence, the biggest area
15. A 16 by 7 with a 5 by 7 removed — area
16. Distance from (7,2) to (7,10)
17. (3,2), (3,8), (7,8), (7,2) — the area

CHALLENGE AND THE ERROR JOURNAL ARE ON THE BACK

Challenge

NOT GRADED · REASON IT OUT

19. A triangle with 90° and 30° — the third angle

20. A 20 by 15 room with a 4 by 5 alcove added — total area

21. A patrol (1,1),(7,1),(7,5),(1,5) — the area enclosed

22. A triangle with 90° and 45° — the third angle

23. A 30 by 20 room with a 6 by 5 alcove added — total area

24. A patrol (2,2),(9,2),(9,7),(2,7) — the area enclosed

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

Week 5 Answers

Mark the reasoning, not the handwriting.

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

5.1 · Map a Planet

OUT OF 21

Warm-Up · 1–11 · 2 | 7 | 4 | 5 | 0,0 | 3,8 | 3 | 9 | 5 | 7 | 5,5

Core · 12–21 · 12 | 36 | 24 | 5,2 | 11 | 13 | 64 | 32 | 6,4 | 13

Challenge · 22–25 · 50 | 27 | 72 | 44

5.2 · Sailing Directions

OUT OF 22

Warm-Up · 1–12 · 5,0 | 5,3 | 8 | 4,6 | 6,4 | 7 | 7,0 | 7,2 | 9 | 5,8 | 8,5 | 11

Core · 13–22 · 10 | 13 | 12 | 7 | 20 | 13 | 15 | 16 | 8 | 28

Challenge · 23–26 · 20 | 24 | 24 | 35

5.3 · Mission Review

OUT OF 18

Warm-Up · 1–10 · 20 | 25 | 4 | 5 | 180 | 24 | 36 | 3 | 2 | 360

Core · 11–18 · 8 | 36 | 15 | 6 | 9 | 64 | 24 | 8

Challenge · 19–26 · 90 | 60 | 84 | 120 | 180 | 70 | 160 | 108

Thu · Error Journal Sweep

OUT OF 22

Warm-Up · 1–12 · 24 | 24 | 360 | 6 | 5 | 2 | 32 | 36 | 360 | 10 | 8 | 4

Core · 13–22 · 42 | 9 | 120 | 18 | yes | 68 | 12 | 145 | 18 | yes

Challenge · 23–26 · 100 | 60 | 225 | 45

Fri · Mission 06 Test

OUT OF 20

Warm-Up · 1–4 · 24 | 20 | 26 | 12

Core · 5–20 · 96 | 4 | 25 | 45 | 7 | 20 | 5 | 60 | 144 | 5 | 100 | 77 | 8 | 24 | 7 | 45

Challenge · 21–24 · 320 | 24 | 630 | 35